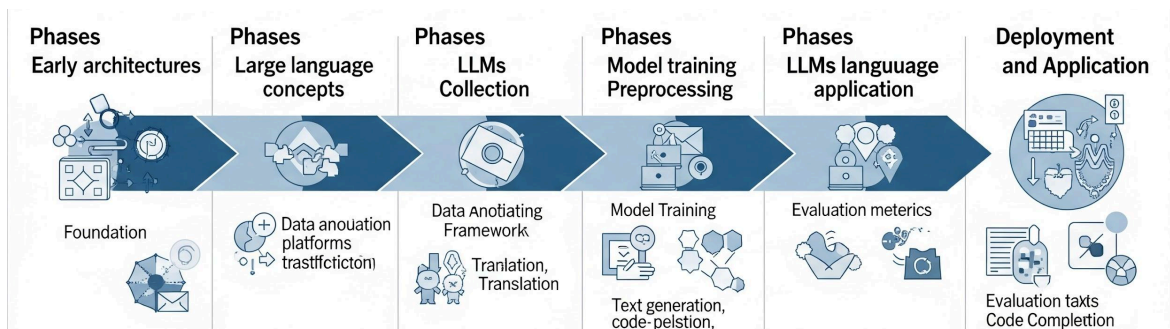




60-Day LLM Learning Roadmap

This roadmap outlines a comprehensive 60-day plan for beginners to learn about Large Language Models (LLMs) and related technologies, culminating in the deployment of several practical projects.

Overall Learning Path

- Learn Python Basics (Day 1–15)
- Learn HTTP + APIs (requests)
- Learn Flask for backend
- Learn OpenAI + Prompting (Day 16–20)
- Learn LangChain Basics (Day 21–25)
- Build FAQ Chatbot (Day 26–30)
- Learn RAG + Vector DB (Day 31–35)
- Build PDF QnA Bot (Day 36–40)
- Learn LangGraph (Day 41–45)
- Build Planner Agent (Day 46–50)
- Final Project + Deployment (Day 51–60)
- Done

Day-by-Day Plan (Low-Level, Beginner Friendly)

Phase 1: Python (Day 1–15)

Days	Task
1–3	Python setup, print, input, variables, datatypes

Days	Task
4–6	Lists, loops, functions, if/else
7–8	Dictionaries, error handling
9–10	Classes, objects, modules
11–12	APIs with requests, call OpenAI/Weather APIs
13–14	Flask Basics: Routes, GET/POST
15	Build mini Flask app (e.g., weather search)



Phase 2: LLM & OpenAI API (Day 16–20)

Days	Task
16	What is LLM, GPT, how it works
17	OpenAI API: setup, call completion endpoint
18	Prompting: zero-shot, few-shot, system role
19	Create chatbot using prompt templates
20	Wrap up + small project: AI intro bot



Phase 3: LangChain (Day 21–25)

Days	Task
21	Install LangChain, basic structure
22	LLMChain, PromptTemplate
23	Chains: SimpleSequentialChain

Days	Task
24	Tools & memory in LangChain
25	Agent basics + tool usage

Project 1 (Day 26–30): FAQ Chatbot

Days	Task
26	Create chatbot skeleton with Flask/Streamlit
27–28	Add LangChain LLMChain + PromptTemplate
29	Add memory for multi-turn chat
30	Test, finalize, and deploy

Phase 4: RAG + Vector DB (Day 31–35)

Days	Task
31	What is RAG? Overview
32	Embeddings: OpenAI/FAISS
33	Load documents (PDF, txt)
34	Store and search using FAISS
35	Combine with LLM to answer questions

Project 2 (Day 36–40): PDF QnA Chatbot

Days	Task
36–37	Build ingestion pipeline (PDF → chunks → vectors)

Days	Task
38	Build search + LLM answer pipeline
39	Add chat UI using Streamlit
40	Test + deploy

Phase 5: LangGraph (Day 41–45)

Days	Task
41	LangGraph basics: node, edge, state
42	Create a simple graph with 3 steps
43	Add branching (if/else) logic
44	Use tools inside graph
45	Test graph with memory

Project 3 (Day 46–50): Planner-Executor Agent

Days	Task
46	Planner node: break input into tasks
47	Executor nodes: execute one-by-one
48	Add memory + stream response
49	Error handling, fallback steps
50	Wrap up and deploy

Phase 6: Final Polish (Day 51–60)

Days	Task
51	Add logging, retry mechanism

Days	Task
52	Evaluate LLM answers
53	Improve prompts using feedback
54	Clean code, add README
55	Upload to GitHub
56	Optional tool agent (e.g., search)
57	Collect user feedback
58	Build portfolio site/Notion
59	Create video walkthrough (optional)
60	DONE  – Review everything